KHADEEJA ABBAS

+1 (587)-436-3492 | khadeeja.abbas@ucalgary.ca | AB, Canada | Portfolio | GitHub | LinkedIn

Interests: Embedded Systems | Vehicle Software | Autonomous Vehicles | **Available**: Immediately

SKILLS

- **Programming Languages:** C, C#, Python, Swift, Java, Assembly
- **Technologies:** STM32, Bare-Metal Programming, FreeRTOS, ARM Cortex-M, CAN/FDCAN, ADC, Git
- **4+ years** of hands-on embedded systems development for a competitive solar vehicle

EVIDENCE OF EXCELLENCE

Solar Car Team - GitHub

Embedded Software Developer → Project Lead → Technical Lead Oct. 2022 – July 2026

- **Developed** bare-metal STM32 firmware in C for the Battery Protection System (BPS), implementing high-voltage contactor control, precharge sequencing, timer-driven control logic, ADC sampling, GPIO interfaces, and inter-board CAN communication.
- **Restored vehicle operability** under **extreme time pressure** at Formula Sun Grand Prix (FSGP) 2025 as the **sole embedded lead on-site**, adapting STM32 firmware to evolving hardware revisions while validating Battery Protection System state transitions and working alongside the electrical team.
- **Validated** Battery Protection System firmware through bench testing and CAN message analysis, verifying inter-board communication, firmware state transitions, and contactor sequencing during hardware bring-up.
- **Reduced development time by 2 months** by identifying communication gaps between electrical and embedded sub-teams, improving cross-team coordination and keeping development aligned.
- **Selected** as Technical Lead for Formula Sun Grand Prix 2026, organizing mock scrutineering, assigning ownership of regulatory requirements, and coordinating completion of critical tasks across mechanical, electrical, and embedded software teams.
- **Expanded Battery Protection System documentation by 40%**, creating technical references and onboarding guides that accelerated knowledge transfer for new embedded developers.

Personal Diagnostics Program

Undergraduate Researcher Sept. 2025 – Present

- Led a research project in one of **the institute's top-funded laboratories**, with work expected to culminate in **two peer-reviewed publications**.
- **Developed** Python-based methods for automated myocardial fibrosis segmentation using a clinical dataset of 500+ cardiac MRI patients, supporting AI-driven cardiovascular research.

PROJECTS

Portfolio

- **Designed and developed** an interactive 3D portfolio website using Three.js, JavaScript, Blender, and GSAP, featuring custom game mechanics, collision detection, inventory and shop systems, responsive controls, and interactive project showcases, transforming a traditional portfolio into an engaging technical experience.

Data Generation for Deep Learning with Unity

- **Generated** a synthetic dataset of 10,000+ aerial wheat images in Unity and developed a Python U-Net segmentation model to improve agricultural computer vision.

Tuber!

- **Built** a Python computer vision application for medication identification, training an image classification model with Apple Create ML and integrating image processing for automated recognition.

Can You Handle Milo?

- **Developed** a Tamagotchi-style game in C on an ARM Cortex-M3 development board, implementing interrupt-driven input, LCD graphics, and finite-state game logic.

EDUCATION

Honours Bachelor of Science, Computer Science Calgary, AB

University of Calgary Graduated with First Class Honours (2026)